DESIGN SPECIFICATION | REVISION 8

Edmonds–Karp Maximum Flow

Interactive svg-dom-graph teaching demo · 3 November 2026 · Implementation gate: Phase 0

Revision 8: three final interaction and build refinements. Preserve V7's validated five golden traces, deterministic generator and mutation coverage. **Every logical playback stop** increments playback_epoch, not just a failed switch; the reverse-arc mutation hook exists **only in test builds**; and deliberate aria-disabled boundary presses give brief, polite feedback after successful dirty repair. No change to solver, event ordering or fixtures.

1. Purpose, baseline and non-goals

Teach shortest augmenting paths in a directed residual network, bottleneck selection, flow conservation, reversal/cancellation of previous flow and the final max-flow/min-cut certificate. The implementation tests scene layering, graph identity, SVG routing, diff rendering, accessibility, deterministic step tracing, playback and atomic scene replacement.

Reuse: Dijkstra as the baseline for directed labelled edges and focus; N-Queens for demo-local SVG layers, transactional control changes, trace budgets and coalesced render scheduling. Use a **single demo-owned control bar**, not the library selection toolbar.

Out of scope: editing arbitrary graph layouts, min-cost flow, interactive graph input, a physics engine and benchmarks. The pure solver accepts small multigraphs; the UI offers two pinned teaching fixtures.

2. Algorithm and input contract

2.1 Validated graph model

Vertices have unique IDs and original directed edges have unique IDs e0…eN in declaration order. The source and sink exist and differ. An original edge e=(u,v,c) has c: u32, flow f: u64, forward residual c−f and reverse residual f. Traversal of a reverse residual transition **subtracts** from that original edge's flow. Parallel and antiparallel edges are legal; self-loops are rejected.

Property	Fixed contract
Size	$2 \le \lvert V\rvert \le 12$; $0 \le \lvert E\rvert \le 48$; two or more distinct vertices even when no edges exist.
Capacities	u32 incl. 0; flow, bottleneck, total, cut sum and residual in u64. Zero edges remain visible as 0/0 but are not examined by BFS.
Arithmetic	Maximum original capacity sum $48 \times (2^{32}-1) = 206{,}158{,}430{,}160 <$ u64::MAX. No public ArithmeticOverflow; retain debug/checked internal arithmetic.
Budget	20,000 snapshots **including Initial**. Before append #limit+1 return EventBudgetExceeded { limit, attempted }; never expose a partial trace.

2.2 Complete typed input errors

TooFewVertices { count, min: 2 }; **TooManyVertices { count, max: 12 }**; **TooManyEdges { count, max: 48 }**; **DuplicateVertexId { vertex_id }**; **DuplicateEdgeId { edge_id }**; **UnknownSource { source_id }**; **UnknownSink { sink_id }**; **SameSourceAndSink { vertex_id }**; **UnknownEndpoint { edge_id, vertex_id, end: From|To }**; **SelfLoop { edge_id, vertex_id }**; and the trace-budget error. Capabilities expressed by u32 cannot produce a negative-capacity error.

2.2 (continued) — deterministic validation precedence

Validation precedence: vertex count/IDs; source/sink existence and distinction; edge count/IDs; **all edge endpoint checks before any self-loop check**; self-loops; trace budget. Within the endpoint pass, use declaration order, From before To. Thus a self-loop on an unknown vertex reports UnknownEndpoint, never SelfLoop. Test every variant and precedence; empty vertices report TooFewVertices, not UnknownSource.

2.3 Deterministic BFS, event emission and invariants

For each dequeued vertex, gather all incident original edges in the orientation that **leaves that vertex**, discard transitions with residual = 0, and sort the remaining transitions by **original edge ID ascending within that vertex**. No second direction tie-break. Emit ArcExamined for every eligible transition encountered, including a destination already discovered. Emit VertexDiscovered only on first visit and record (vertex, edge ID, direction) predecessor. Stop the successful BFS as soon as t is first discovered.

Each search emits SearchStarted; each dequeue emits VertexDequeued. A successful BFS emits PathFound, BottleneckDetermined and one atomic FlowAugmented; then start the next search. An exhausted search emits SearchExhausted followed by MinCutShown. The final BFS discovered set is **exactly S**; T=V\S and the cut counts original edges S→T, not T→S.

After each augmentation verify $0 \le f \le c$, intermediate-vertex conservation, source net outflow = sink net inflow = total_flow, correct paired residuals and nondecreasing total flow. A reverse arc may decrease an individual edge flow while increasing total flow.

3. Locked teaching fixtures and golden results

3.1 Canonical network — max flow 23

ID	From	To	Cap.	ID	From	To	Cap.
e0	s	v1	16	e5	v3	v2	9
e1	s	v2	13	e6	v2	v4	14
e2	v1	v2	10	e7	v4	v3	7
e3	v2	v1	4	e8	v3	t	20
e4	v1	v3	12	e9	v4	t	4

Path sequence: s→v1→v3→t, +12 (total 12); s→v2→v4→t, +4 (total 16); s→v2→v4→v3→t, +7 (total 23).

Final flows e0..e9: [12, 11, 0, 0, 12, 0, 11, 7, 19, 4]. **Cut:** S={s,v1,v2,v4}, T={v3,t}; crossing e4=12, e7=7, e9=4 → capacity 23. e5 is T→S and is excluded. Exactly **87 snapshots**: Initial + 86 events, indices 0..86.

Coordinates: s=(80,200), v1=(260,90), v2=(260,310), v3=(470,90), v4=(470,310), t=(660,200); viewBox 0 0 740 400.

3.2 Mandatory reverse-cancellation fixture — max flow 3

ID	From	To	Cap.	ID	From	To	Cap.
e0	v1	v3	2	e4	s	v1	2
e1	v3	t	2	e5	s	v4	1
e2	v4	v3	1	e6	v2	t	2
e3	v1	v2	2				

Path 1: s→v1→v3→t (e4 F, e0 F, e1 F), delta 2, total 2. **Path 2:** s→v4→v3→v1→v2→t (e5 F, e2 F, e0 R, e3 F, e6 F), delta 1, total 3. The e0 reverse hop changes e0 flow **2→1**. This is tie-break-dependent and must be visibly and textually explicit.

Final flows e0..e6: [1, 2, 1, 1, 2, 1, 1]. S={s}, T={v1,v2,v3,v4,t}; cut e4=2 + e5=1 → 3. Exactly **44 snapshots**: Initial + 43 events, indices 0..43.

3.2 (continued) — planar fixed layout

Positions: s=(80,200), v1=(230,80), v2=(490,80), v3=(490,320), v4=(230,320), t=(650,200); viewBox 0 0 740 400. This fixture's seven original edges have no interior crossings at the specified coordinates.

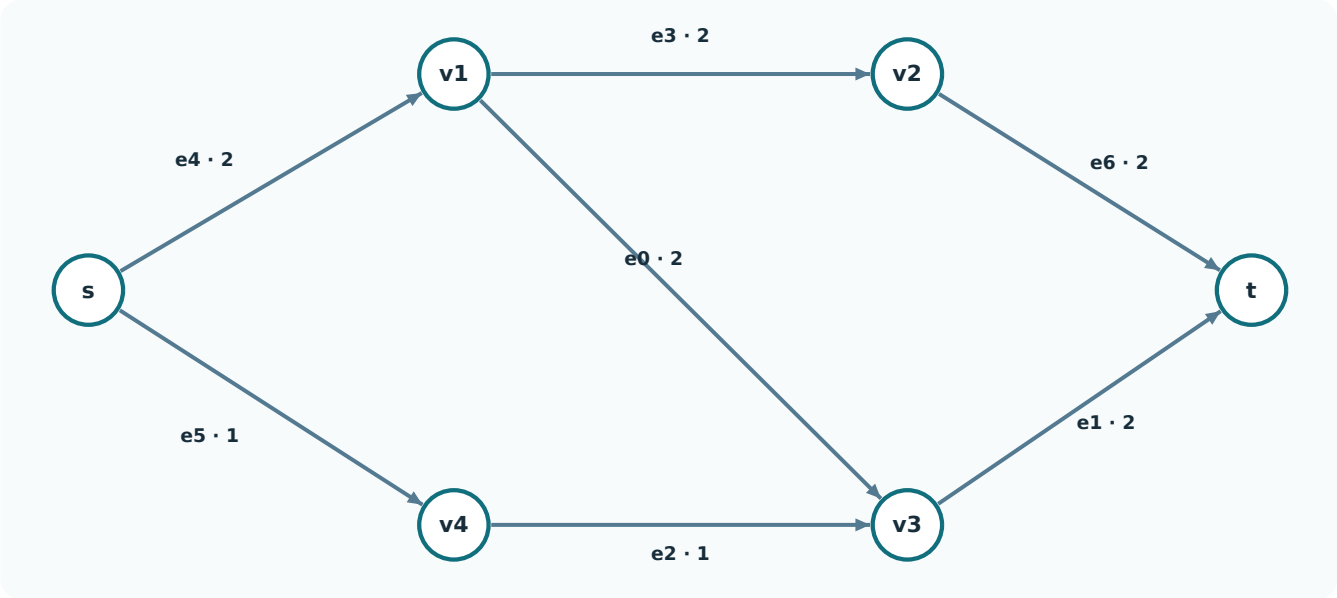


Figure 1. Pinned reverse fixture original-edge layout; the v3→v1 step is the residual reverse of e0, not an extra original edge.

3.3 Golden event histogram (snapshots include Initial)

Event variant	Canonical	Reverse
Initial	1	1
SearchStarted	4	3
VertexDequeued	17	10
ArcExamined	36	12
VertexDiscovered	18	10
PathFound	3	2
BottleneckDetermined	3	2
FlowAugmented	3	2
SearchExhausted	1	1
MinCutShown	1	1
Total snapshots	**87**	**44**

Budget-injection golden boundaries: reverse limit 43 rejects attempted snapshot 44; limit 44 succeeds. Canonical limit 86 rejects attempted 87; 87 succeeds. Inject limit into the pure trace builder for tests, leaving the production default at 20,000.

4. Additional pinned solver fixtures

The following compact graphs are **unit-test-only** fixtures, not added to the two-option UI. Golden sequences use exactly the §2.3 event grammar and detect wrong-but-conserving implementations that would pass invariant-only tests.

4.1 Unreachable sink, reachable set and zero-capacity edge

Vertices [s,a,b,t]. Edges in declaration order: e0 s→a (2); e1 a→b (1); e2 b→a (1); e3 b→t (0). No augmenting path. Total flow 0; all original flows 0; S={s,a,b}, T={t}, S→T crossing e3 capacity 0; min cut 0. The zero-capacity edge e3 is **never ArcExamined**.

Index	Event (exact order)
0	Initial
1	SearchStarted
2	VertexDequeued(s)
3	ArcExamined(e0 F, s→a, unvisited)
4	VertexDiscovered(a)
5	VertexDequeued(a)
6	ArcExamined(e1 F, a→b, unvisited)
7	VertexDiscovered(b)
8	VertexDequeued(b)
9	ArcExamined(e2 F, b→a, already discovered)
10	SearchExhausted
11	MinCutShown — S={s,a,b}, cut 0

Exactly 12 snapshots; the final failed search's discovered set must equal S. No PathFound, BottleneckDetermined or FlowAugmented events.

4.2 Zero capacity alone: five-snapshot terminal trace

Vertices [s,t], e0 s→t with capacity 0. Golden order: **Initial → SearchStarted → VertexDequeued(s) → SearchExhausted → MinCutShown**. Exactly 5 snapshots, flow 0, S={s}, cut capacity 0. The presenter shows e0 as 0/0, while the solver emits no ArcExamined for e0.

4.3 Parallel edges: distinct IDs and deterministic tie-break

Vertices [s,a,t]; e0 s→a (1), e1 s→a (2), e2 a→t (3), in that declaration order. Search 1 examines e0 F first, discovers a, then examines e1 F as already discovered; path e0 F/e2 F, delta 1. Search 2 examines e1 F from s; at a it examines e0 R (back to discovered s) before e2 F; path e1 F/e2 F, delta 2.

Final flows [1,2,3]; total 3; S={s}; cut e0 + e1 = 3. **Exactly 27 snapshots**, 26 events: Initial 1, SearchStarted 3, VertexDequeued 5, ArcExamined 6, VertexDiscovered 4, PathFound/BottleneckDetermined/FlowAugmented 2 each, SearchExhausted 1, MinCutShown 1.

Exact event-ID sequence: 0 Initial; 1 SearchStarted; 2 Dequeued(s); 3 Examined(e0 F); 4 Discovered(a); 5 Examined(e1 F, seen); 6 Dequeued(a); 7 Examined(e2 F); 8 Discovered(t); 9 PathFound; 10 Bottleneck; 11 Augmented; 12 SearchStarted; 13 Dequeued(s); 14 Examined(e1 F); 15 Discovered(a); 16 Dequeued(a); 17 Examined(e0 R, seen); 18 Examined(e2 F); 19 Discovered(t); 20 PathFound; 21 Bottleneck; 22 Augmented; 23 SearchStarted; 24 Dequeued(s); 25 SearchExhausted; 26 MinCutShown.

5. Immutable trace, view state and presentation

Precompute one complete immutable solver trace before first render. Indices are contiguous; “Step i of M” means snapshot i where 0 ≤ i ≤ M, M=snapshot_count−1. Canonical M=86, reverse M=43. The slider uses min=0/max=M. There are no hidden pseudo-events or alternative compact indices.

Trace snapshot { index, event, flows, total_flow, bfs, path, bottleneck, augmentation, final_cut }
Augmentation { path: Vec<(edge_id, direction)>, delta, flows_before, flows_after, total_before, total_after }
CommittedView { fixture_key, index, focused_target, roving_target, show_residual, show_graphical_queue, detailed_examinations, speed }
Controller { generation, pending_seek, frame_handle, tick_handle, playback_epoch, dirty, current_scene, lifecycle_token }
Derived playing = a valid playback tick is pending for the current playback_epoch; it is **not stored** in CommittedView.

Each snapshot is standalone and semantic: BFS includes queue order, discovered set, predecessor map, examined original edge ID, orientation and visitation outcome; the augmentation snapshot carries complete before/after vectors. View focus, generation, DOM handles, scheduler handles and toggle preferences never enter pure solver snapshots or semantic snapshot equality.

Compact examinations change explanatory verbosity only. Even with details hidden, the current-event line always changes for every index (index, event kind, inspected edge/direction/endpoints and discovered/ignored outcome when applicable). The textual queue remains visible regardless of the graphical queue toggle.

6. SVG layers, identities and routing

6.1 Element identity and stable selectors

Use a persistent original-edge layer plus a demo-local residual layer containing positive forward/reverse transitions. Original labels always show flow/capacity, including 0/0. Residual lanes are keyed by (original edge ID, F|R) and do not renumber as residual capacity reaches zero. Distinguish orientation by arrowheads, dashed lane style, offset labels and matching text; color is never the only cue.

Every SVG **id** includes instance + activation generation + fixture key: e.g. ek-<instance>-g<generation>-reverse-edge-e0. **Activation generation, trace generation and Controller.generation are the same single counter.** Staged scene IDs use current generation + 1; the counter advances **only** on a successful committed switch. Discard failed stages and their listeners before reusing that candidate value. Use unique IDs for aria-labelledby and staged/live collision prevention; never retain cross-fixture edge focus.

On **every original edge** and its hit target, expose stable data-ek-edge="e0"; on vertices expose data-ek-vertex="v1". Residual paths also carry data-ek-edge of the parent and data-ek-direction="f"|"r". Root has data-ek-fixture="canonical"|"reverse". Tests scope data selectors to the active fixture root, not ephemeral SVG IDs; aria still uses unique IDs.

6.2 Fixed geometry and synthetic six-curve visual stress case

Stable offsets depend on unordered endpoint pair, original edge ID and orientation, with a separate residual lane family. Original canonical antiparallel e2/e3 occupy opposite curved sides. **Visual routing guarantees apply only to the two fixed UI layouts and the pinned six-curve stress snapshot**, not any arbitrary 48-edge solver input.

Pinned presenter-only state	Required visible geometry
Canonical graph; set e2 flow = 3 of 10 and e3 flow = 2 of 4; leave other original-edge flows 0.	With **show_residual=true**: two original v1↔v2 curves e2 3/10 and e3 2/4 plus four residual lanes e2 F=7, e2 R=3, e3 F=2, e3 R=2: **six distinct curves**. Synthetic flow need not conserve; presenter must not assert solver invariants. With false: two originals. Check no curve/label/arrow overlap at 100% and 150%.

6.3 Diff commits and dirty repair

Render each target (snapshot, view) by diff against the last fully committed presentation. Update only changed SVG geometry/classes, original/residual labels, accessibility names, queue/event/path text, final cut and control attributes. Keep a from-scratch presenter for dirty repair and equivalence testing.

A required DOM-write failure **does not commit** the index or any view-only toggle/focus change. Best-effort revert touched live controls, including aria-pressed, to the committed values and set dirty even if revert appears complete. A failed fixture switch is the exception for playback controls: old fixture/view/index stay, but the committed target for Play/Pause is **paused** and its playback_epoch advances through stop_playback(). On the **next action, including a boundary no-op or a same-fixture selector choice**, first fully re-render the prior committed Snapshot+View **and live controls**; only then process that action. If repair fails, abort the action and keep dirty. No partially changed DOM is considered committed.

Verify at **DOM level**: for every sequential diff through the full trace, compare the normalized live SVG/HTML DOM with a newly rendered DOM for that exact same snapshot and view. Repeat across random seek orders. Normalize only ephemeral element IDs and legitimately generated tokens; compare paths, labels, aria, queue text and cut state.

7. Fixture switching and control transactions

7.1 Selector and atomic scene replacement

Persistent labelled selector with exactly **Canonical (max flow 23)** (default) and **Reverse cancellation (max flow 3)**. Same-fixture selection is a no-op after dirty repair. Successful switch resets to Step 0, calls the common **stop_playback()** routine, cancels pending seek, increments the **single** generation counter exactly once, clears edge-specific informational focus, sets roving target to s, and preserves view toggles/speed.

Prepare off-DOM: validate/build the trace, fully render the candidate graph *and all dynamic text panels* into one detached replaceable subtree, prepare the target tabindex and namespaced IDs at generation+1. Save the **committed** selector value, controls, active children, focus, snapshot, tick/seek handles and generation; perform a root-swap injection gate **before any live write**. An unsuccessful detached preparation leaves old graph/index/generation intact but **does pause playback** (call stop_playback(), update Play control to paused or mark dirty if that write fails). **There is no dry-run for a live DOM setter.**

Journal and attempt: apply the explicitly enumerated persistent-control writes in §7.1a while keeping old children attached. Record old values for each property/attribute. Inject a throw at **each individual write**, including a failure after preceding writes succeeded. If any write fails, restore touched controls in reverse order (best effort), discard the candidate, leave committed model/index/generation unchanged, **pause playback and invalidate pending work**, and set dirty. Test a failed restore separately; the next action must repair from committed state before proceeding.

Root replacement and commit: after live writes succeed, attempt actual host child replacement; injection may throw, including after partial replacement. On failure best-effort restore old subtree and controls, retain the old model/index/generation and mark dirty. Only **after successful replacement** commit the candidate and increment generation once; invalidate old callbacks, stop playback and transfer graph focus safely. **Every failed switch invokes stop_playback()**, even detached-build/preflight failures: cancel/clear the tick and advance playback_epoch so late ticks cannot run despite unchanged activation generation. Cancel/invalidate pending seek frames too. Failed writes restore the old fixture's **paused** control state; any failed repair leaves dirty. Detached-only failures need not set dirty if nothing live was touched. Announce only successful changes.

7.1a Live controls changed by a fixture switch

Target outside replaced subtree	Required candidate write; saved old value
Fixture selector	Selected value; a user-changed value is provisional until the switch commits.
Seek slider	Native range: max = new M, value = 0, aria-valuetext = “Step 0 of M”. min remains 0. **Do not write aria-valuemax or aria-valuenow**: native max/value expose them.
Persistent step readout	textContent = “Step 0 of M”; test every text and attribute setter.
Six navigation buttons	At Step 0: aria-disabled=true for Prev, Restart, Previous augmentation; false for Next, Next augmentation, Jump to cut. **No native disabled property.**
Play/Pause control	Step 0: show enabled Play (aria-disabled=false). Preserve old pending tick during fallible staging; stop it on either failed attempt or successful commit.

Use only **aria-disabled** and a visual class for all navigation bounds, never native disabled or pointer-events:none: the focusable button’s click/keyboard handler must still run dirty repair before its boundary no-op. At index 0 exactly the three buttons above are unavailable; Next, Next augmentation, Jump to cut and Play are enabled. Unavailable means an aria-disabled presentation, **not** suppression of user handlers. Derive states for later indices from legal targets. Preserve queue/residual/detail/speed preferences. Stage the dynamic text with the SVG. Journal and fault-test each live write, including aria-disabled and aria-valuetext.

7.2 Controls — fixed navigation and state ownership

Boundary handlers are always executable: aria-disabled communicates unavailable actions but must not stop click/keyboard delivery. Each navigation handler first attempts dirty repair; if repair succeeds it checks the boundary, and then commits an index change or performs a no-op **with one polite boundary message** for a deliberate user press. A repair failure leaves the page dirty, aborts navigation and emits no boundary message. This also preserves keyboard focus at Step M.

Control	Normative behavior
Prev / Next / Restart	Manual action calls stop_playback(); ±1 snapshot index, or index 0. Boundaries remain index no-ops after dirty repair but announce one brief boundary message.
Previous augmentation	Strictly largest FlowAugmented index **less** than current. At/before first augmentation with none preceding: no-op with polite message. Never reselect current augmentation.
Next augmentation	Strictly smallest FlowAugmented index **greater** than current. At/after last augmentation with none following: no-op with polite message. Never reselect current augmentation.
Jump to end/cut	Jump to terminal MinCutShown index M (Go to end). At M: index no-op with polite message. Same seek commit, stop, focus and announcement semantics.
Seek slider	input coalesces ≤1 render per animation frame, with generation-tagged pending target. Only committed Step i/slider state displayed; change/release flushes final target and announces once.
Graphical queue toggle	Show/hide SVG queue badges/order marks **only**. The accessible ordered textual queue is always present. A failed toggle preserves its committed aria-pressed.
Other view toggles	Residual and examination detail change presentation only; focus changes informational emphasis only. None changes snapshot indices or solver state.

7.3 Speed, playback and scheduler failures

Selection	Tick interval	Status
0.5×	800 ms	slow
1×	400 ms	normal
2×	200 ms	default
4×	100 ms	upper bound

Intervals define a bounded playback rate (**2×/200 ms default**: about 17 seconds for 86 canonical transitions). Changing speed affects subsequent ticks but not snapshots or indices; if playing, invalidate the old tick epoch **before** scheduling its replacement. Use an injectable Scheduler: schedule_once(delay, cb), cancel(handle), request_frame(cb), cancel_frame(handle), each scheduling request being able to fail in tests. Virtual time drives all playback unit tests.

One source of truth: playing is derived only from an outstanding valid tick_handle for the current playback_epoch, not copied into CommittedView. Define **stop_playback()**: cancel/clear the pending tick, then **unconditionally increment playback_epoch on every logical stop**, even when cancel fails or no handle exists. Call it on Pause, every manual navigation/seek, an active speed change (before scheduling a new tick), the terminal step, render/scheduler failure, every failed *or* successful fixture switch, and teardown. A Play from paused schedules at most one tick with the current epoch. When a tick fires normally, consume its handle and commit its step; schedule another tick **only after successful rendering**. Consuming an intermediate tick is **not** a logical stop and does not increment the epoch. A reschedule failure stops and reports a nonblocking error.

A failed animation-frame request for a pending seek clears its pending handle/target and attempts the latest seek **synchronously once**. Commit and optionally warn on success; on required DOM failure retain the old index, set dirty and announce nothing. Slider release/change flushes once and suppresses duplicate announcements. Late frame/tick callbacks carry generation + lifecycle tokens; ticks additionally capture playback_epoch at scheduling and no-op on **any** mismatch, including ordinary Pause → Play, manual navigation, speed replacement, terminal, failed switch or teardown. All failures leave the Play/Pause control reflecting the committed paused state after repair.

8. Keyboard access, semantics and motion

Use **roving tabindex**: one graph Tab stop, deterministic target order [s, v1, v2, v3, v4, t, original e0..eN]. Exactly one graph target has tabindex=0, the rest −1. Canonical: 16 roving targets; reverse: 13. **Left and Up both move to the preceding list item; Right and Down both move to the next list item** (wrapping). Directions are **list order, not spatial neighbors**. Home/End select first/last. Only intercept keys while graph focus is active; leave native selector/slider controls alone.

Vertices and original edges are focusable targets. Residual transitions are **not** separate keyboard targets; clicking a residual lane selects/informationally focuses its parent original edge identified by data-ek-edge, without changing solver index. Graph pointer clicks are otherwise informational focus changes only. Focus transfer after fixture swap is transactional and never leaves focus on a removed node.

Original edge accessible names include edge ID, endpoints, flow/capacity, forward residual and reverse residual. The residual text specifies “forward of e0” or “reverse of e0”, its traversal endpoints and capacity, especially when original antiparallel edges coexist.

Text panel always shows **Step i of M**, phase, ordered BFS queue, examined edge/direction/outcome, visited vertices, current path, bottleneck, per-edge delta, running flow and final partition/cut. The text queue never disappears when the **graphical** queue toggle is off. Compact examination mode must visibly change its event line even on otherwise visually identical steps.

Announce phase+step for successful manual Prev/Next/jump commits; slider movement announces once on completed change/release, not each input. Playback announces only PathFound, FlowAugmented and MinCutShown. **Exception for deliberate boundary presses:** after dirty repair succeeds, emit **one aria-live=polite message** without changing index, focus or snapshot: Prev/Restart at 0: “Already at the first step”; Next/Jump to cut at M: “Already at the last step”; Previous augmentation with none: “No earlier augmentation”; Next augmentation with none: “No later augmentation”. A handler invoked by Enter/Space behaves like click. No boundary message on repair failure, stale callbacks, automatic playback, abandoned scene swaps or programmatic control updates. These explicit no-op announcements do not count as snapshot commits.

If a decorative flow pulse/highlight animation exists, disable it under prefers-reduced-motion while retaining static arrow, line and text emphasis. No node motion or physics; all meanings have non-color and textual equivalents.

9. Verification matrix (mandatory)

Area	Assertions and fixtures
Canonical solver	Exact three paths, bottlenecks [12,4,7], flows [12,11,0,0,12,0,11,7,19,4], cut 23, 87 snapshots and canonical 9-variant event histogram (§3).
Reverse cancellation	Exact paths, second e0 R and 2→1 change, final [1,2,1,1,2,1,1], S={s}, cut e4+e5=3, 44 snapshots and event histogram.
Disconnected sink	Four-vertex fixture §4.1: exact 12-step event sequence, no paths, S={s,a,b}, max/min cut 0; e3 cap0 never ArcExamined.
Zero capacity	Two-vertex fixture §4.2: exact five snapshots, 0/0 visible, no eligible/examined residual arcs, final S={s}, cut 0.
Parallel edges	Three-vertex fixture §4.3: exact 27 snapshots, e0-before-e1 initial BFS examination, e1-before-e0 R next search, paths/deltas [1,2], final [1,2,3] and cut 3.
Broad independent oracle	Keep V5: exactly 500 seed-0xEDCA_2026_1103_0001 multigraphs, 2..=8 vertices, 0..=16 edges, capacities 0..=7, distinct s/t, parallel/antiparallel/zero permitted, no self-loops. Enumerate all 2^(V−2) cuts (≤64 each) and compare correct solver value.
Layered oracle / coverage	Appendix A **500** graphs: exact first graph, 6,067 total edges and capacity sum 12,188; 0 loops; all max flows equal min cuts (≤64 partitions each); nonzero=500, reverse-used≥10 (baseline 24).
Negative-control meta-test	Use a compile-time **#[cfg(test)]** allow_reverse parameter/branch within the shared BFS; no production flag. Enabled control matches normal entrypoint flow/paths on all 500 (plumbing check); disabled mismatches independent cut on ≥10 (baseline 24: substantive evidence).
Final search & invariants	The final BFS discovered set equals S exactly; cut uses only original S→T edges. Bounds, conservation, reverse residual arithmetic and total after **every** augmentation.
Typed validation	Exercise each typed error and precedence, including unknown self-loop endpoint: UnknownEndpoint wins over SelfLoop, with From checked before To; source/sink absent and TooFewVertices precedence. No public overflow error.
Trace budget	Inject 43/44 reverse and 86/87 canonical budgets; reject attempted=44/87 and publish no partial trace; production cap 20,000 including Initial.
Trace independence	Seek any snapshot from scratch; same IDs/index totals with compact on/off and queue graphics on/off; compact inspected-arc text changes at every step.
Jump navigation	Next/previous augmentation use strict greater/less indices even from an augmentation; no index change past boundary. Jump-to-cut lands on M. Successful seek emits one announcement; **each deliberate boundary press** emits its pinned polite text after repair.

9. Verification matrix (continued)

Area	Assertions and fixtures
Fixture transaction	Inject detached failures, each live selector/range/value/text/aria-disabled write, restoration and actual replacement. Assert old model/index/generation; failed switch **pauses** (tick cancelled, playback_epoch changed), no stale tick, old paused UI restored or dirty; boundary action repairs first.
Stable vs unique IDs	One generation counter for activation/trace/controller; candidate IDs use g+1, discarded candidates do not advance it, successful commits advance once. Unique staged/live SVG IDs and stable scoped data-ek-* locators across switches.
SVG/rendering	Two original fixtures at base/150% zoom. Synthetic e2=3, e3=2 with **show_residual=true**: exactly six distinct non-colliding v1/v2 curves, residuals 7,3,2,2 and original labels 3/10,2/4. With false: just two originals. No conservation assertion in presenter.
DOM-level diff	At each index compare normalized complete DOM after sequential diff against from-scratch DOM of same (Snapshot, View), including paths, labels, aria, text and cut. Repeat random-order seeks; injection for node, edge, label and aria writes.
View write failure	Toggle, focus, control-aria failures do not commit view/index. aria-disabled boundary click and Enter/Space trigger repair first, then one polite message; failed repair is silent and stays dirty. Assert pinned message for **all six** boundary cases. No native disabled.
Virtual-clock playback	800/400/200/100 ms, default 2×. Stop_playback increments epoch on Pause, manual seek/jump, active speed change, terminal, render/schedule failure, each switch outcome and teardown. Late callback from Pause → immediate Play must be rejected after a new tick is queued.
Seek/frame failures	Coalesce slider input ≤1 frame; failed request_frame invokes one synchronous latest-target render; on DOM failure index unchanged + dirty; release announces once with no duplicate change/pointerup.
Generation/teardown	Old-epoch ticks ignored even if host cancellation fails, including Pause → Play and speed changes with unchanged activation generation; stale seek/frame/tick ignored after swap/teardown; handles, observers, test hooks released.
Keyboard/pointer	Canonical 16/reverse 13 roving targets but exactly one Tab stop. Arrow order linear (not spatial); residual click chooses original parent edge; residual never roving focus target; focus across replacement.
Accessibility & motion	Edge IDs and both residual directions in names; text queue stays visible despite graphical toggle; selective live announcements; no animation with reduced-motion; non-color cues.

10. Delivery phases and acceptance

Phase	Required exit condition
0 — integration audit	Check Dijkstra edge/label/focus APIs, N-Queens layers and dirty-restore precedent; identify all live control setters and real root-replacement failure seam, deterministic rendering and CDP DOM/keyboard seams. No algorithm/test decision deferred.
1 — solver	Typed validation, five goldens, budget; 500 broad + 500 layered graphs, pinned edge/capacity checksums and first graph; ≥10 reversal cases and mutant failures; enabled-control path and flow agree on all 500.
2 — presenter	Both fixed layouts, separate original/residual layers, six-curve test snapshot, stable data selectors + namespaced IDs, accessibility text, diff rendering vs fresh DOM.
3 — controls	Jumps/toggles + polite boundary messages; aria-disabled still actionable; journal/per-setter/root failures; stop_playback epoch rules, stale-tick tests, dirty repair, virtual clock and roving focus.
4 — release verification	Full §9 matrix, zoom and reduced-motion screenshots, teardown, error injection, docs/index updates, security and performance notes.

11. Definition of done and V7 → V8 disposition

Phase 0 / implementation gate. The five golden traces stay immutable (87 / 44 / 12 / 5 / 27 snapshots). The cancellation fixture must visibly subtract flow on e0. The 500-graph layered generator is frozen by call order, first-edge fixture and aggregate checksums. Boundary controls remain actionable with polite no-op messages; **every** playback stop invalidates the old epoch; reverse-disable controls exist only in tests.

What V8 fixes — normative additions

• **Unified playback stop:** stop_playback() invalidates outstanding callbacks by incrementing playback_epoch for *all* stops. The next Play may schedule another tick immediately without allowing a cancelled Pause-era callback through. An intermediate successful playback tick is not a stop.

• **Stale tick acceptance:** explicitly test Pause → Play, manual seek/jump, speed change, completed trace, failed/successful switch, schedule/render failure and teardown. A late callback has the old epoch even if its host cancellation failed.

• **Mutation hook only in tests:** the common BFS code may receive a #[cfg(test)] allow_reverse switch or equivalent test-only parameter, but no runtime flag/branch is built into production. The enabled-mode parity check verifies test plumbing; the disabled-mode mismatch against min-cut is the substantive correctness sentinel.

• **Deliberate boundary press:** after repair, a focusable aria-disabled click or Enter/Space gives exactly one brief aria-live=polite message. No index/snapshot change or commit announcement. No extra speech for aborted repair or background/automatic callbacks.

Preserved V7 commitments

V7’s generator retry order (redraw kind and both endpoints), first graph, 6,067 edges/12,188 capacity checksum, 24/24 baseline, six actionable navigation controls, failed-switch pause, test-control parity and native range write minimization all remain mandatory.

Preserved implementation commitments

Exact event histograms, snapshot model, standalone augmentation before/after flows, per-vertex BFS edge-ID tie-break, validation precedence, five goldens and cut results, two planar UI fixture layouts, synthetic six-lane stress test, immutable index space, DOM-level diff comparison, virtual scheduling and accessible queue/focus contracts are unchanged.

Phase 0 remains limited to matching the specification to repository APIs and test harness seams; it must not silently change the oracle, fixture results or control failure semantics.

Appendix A. Reproducible layered oracle and mutation sentinel

Exactly 500 graphs use a continuous SplitMix64 stream seeded **0xEDCA_2026_1103_0002**. Every next() is unsigned u64; each next()%k consumes one draw. Evaluate tuple expressions **strictly left-to-right**: e.g. (x,R[next()%b],cap()) draws the R index **before** cap(). Vertex IDs 0..n−1, source=0, sink=n−1; edge IDs are declaration indices after Fisher–Yates. cap()=1+(next()%3). Never emit loops; reject and **redraw kind + both endpoints** without capacity on a loop.

Stage	Exact generator rule and PRNG call order
Choose layers	a=2+(next()%2); b=2+(next()%2); n=2+a+b; L=[1..a], R=[a+1..a+b], t=n−1.
Construct backbone	For each x in L add (0,x,cap()); for each y in R add (y,t,cap()); for each x in L add (x,R[next()%b],cap()). Each graph therefore has positive s→t capacity.
Periodic adversarial pair	If graph_index % 4 == 0: append (L[0],L[1],cap()), then (L[1],L[0],cap()). All other graphs omit this forced pair.
Choose length	target = max(8 + (next()%9), current_edge_count), bounded ≤16.
Fill until target	Draw kind=next()%10. 0..1: (0,L[rand(a)]); 2..5: (L[rand(a)],R[rand(b)]); 6..7: (R[rand(b)],t); 8: (L[rand(a)],L[rand(a)]); 9: (R[rand(b)],R[rand(b)]). If u=v, discard **the entire attempt**; repeat from a **new kind=next()%10**, then draw new endpoints. Do **not** draw cap() on rejection. Otherwise append (u,v,cap()).
Declaration order	Fisher–Yates descending i=len(edges)−1..1: j=next()%(i+1); swap edges[i],edges[j]. All loops excluded; duplicate and antiparallel edges permitted.
Independent check	Enumerate all 2^(n−2) subsets of intermediate vertices (≤64), with 0 in S, t outside S; cut=sum capacities of original S→T edges. Compare correct solver max_flow to the minimum cut.
Coverage + mutation	First check n=8/first graph and 6,067-edge/12,188-capacity checksums. Count max_flow>0 (500), reverse-used (≥10), mutant ≠ cut (≥10); enabled test-only shared-core variant matches normal solver flow and paths on all 500.

Generator lock: first graph (n=8), declaration-order edges: [(0,3,3), (3,6,1), (0,2,2), (1,5,1), (1,2,3), (6,7,1), (4,7,2), (1,5,3), (5,7,3), (2,6,1), (2,1,2), (0,2,3), (0,1,1)]. Over all 500 graphs: **total original edges = 6,067; sum of original capacities = 12,188**. Assert the complete first graph and both totals before running any solver or oracle checks.

RNG definition: state=(state+0x9E3779B97F4A7C15) mod 2^64; z=state; z=((z xor (z>>30))*0xBF58476D1CE4E5B9) mod 2^64; z=((z xor (z>>27))*0x94D049BB133111EB) mod 2^64; return z xor (z>>31). Use one state across all 500 graphs; never reseed per graph.

Reproducibility baseline: 24/500 graphs with a reverse augmentation, 24/500 reverse-disabled incorrect values, 500/500 positive flow, 500/500 correct matches to minimum cut. A **#[cfg(test)]** allow_reverse parameter or equivalent is compiled **only** into test builds of the *same* BFS core; the production build has neither its branch nor a runtime flag. With true it matches the normal solver entrypoint on total flow and complete ordered augmentation paths (edge ID, F/R, bottleneck) across all 500: this verifies test plumbing, **not** reverse correctness. With false it skips only reverse transitions; its ≥10 independently measured wrong min-cut values (baseline 24) supply the substantive mutation evidence. No independent mutant implementation.

Appendix B. Fixture-switch fault injection and repair

Use an injectible DOM-writer adapter for each setter; tests must fail at the setter **after** any preceding writes have executed, not merely at a preparatory gate. The adapter must also be able to fail during best-effort restore and during the actual subtree replacement. The graph/model remain separate from live DOM until replacement commits.

Injection family	Expected outcome and assertion
All live control writes	Throw at each §7.1a setter after preceding writes. Old committed fixture/index/generation stay, candidate discarded; revert saved values **except** Play/Pause settles to paused. Cancel tick/invalidate playback_epoch; dirty=true.
Restore itself fails	Throw on setter then reverse-order restore; no model commit. Cancel tick. Repair must restore controls from old model plus **paused** playback and clear dirty only after full success.
Actual root replacement	Throw at actual root replacement after live writes. Restore old children and controls best effort; model/index/generation unchanged, tick paused and invalidated, dirty=true. Retain optional preflight gate.
Repair-before-no-op	While dirty, click and press Enter/Space on an aria-disabled boundary button; handler must run repair before no-op. Also test same-fixture selection and no-op toggle. Repair failure keeps dirty and blocks requested action.
Success after failure	After repair, successful switch increments generation once. At Step 0 aria-disabled true on Prev/Restart/Prev Aug; false on Next/Next Aug/Cut, Play enabled; no native disabled. Old tick stays invalidated.
UI/solver independence	Text and graph stage as one subtree; failed switch never leaks candidate visuals. Selector reflects old fixture; paused control state wins over journal's old Play/Pause; old frames/ticks are stale. IDs reused only after full teardown.

Implementation-order guidance

Repair on entry → prepare detached candidate → preflight injection → journal live controls → attempt actual replacement → commit model/generation → stop_playback()/invalidate pending callbacks → manage focus. On **every successful or failed** switch, use stop_playback(): cancel/clear tick and increment playback_epoch (independent of activation generation); invalidate queued seek/frame. On failure, do this before attempting restore, so a late callback no-ops. Best-effort undo live changes, restoring old fixture/index but **paused** Play state. Dirty=true after any partial live change or root failure; pure detached failure requires no dirty repair. Do not perform required fallible writes after commit.

Note on physical DOM failure: if restoring the old subtree fails, block further actions until dirty repair restores the old scene and all persistent controls. No tick is logically pending after *any* failed switch (playing=false), regardless of whether cancellation of its host callback succeeded. Playback may resume only through an explicit later Play action after successful repair. Test a late old tick while generation is unchanged; playback_epoch must reject it.

Reference baseline: V7 reviewed 3 November 2026. V8 unifies all playback stop invalidation, makes reverse-disabled mutation branching test-build-only and adds deliberate boundary-press feedback. The five goldens, event histograms, oracle stream and solver algorithm are unchanged.